

Making Language Models Follow Instructions

CS60216: Safety Fundamentals for Generative AI

Jan 21st, 2026

2 example models

GPT-3 (2020)

50,257 vocabulary size
2048 context length
175B parameters ✓
Trained on 300B tokens }

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or "GPT-3"	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Table 2.1: Sizes, architectures, and learning hyper-parameters (batch size in tokens and learning rate) of the models which we trained. All models were trained for a total of 300 billion tokens.

Training: (rough order of magnitude to have in mind)

- $O(1,000 - 10,000)$ V100 GPUs
- $O(1)$ month of training
- $O(1-10)$ \$M

LLaMA (2023)

32,000 vocabulary size
2048 context length
65B parameters →
Trained on 1-1.4T tokens →

params	dimension	n heads	n layers	learning rate	batch size	n tokens
6.7B	4096	32	32	$3.0e^{-4}$	4M	1.0T
13.0B	5120	40	40	$3.0e^{-4}$	4M	1.0T
32.5B	6656	52	60	$1.5e^{-4}$	4M	1.4T
65.2B	8192	64	80	$1.5e^{-4}$	4M	1.4T

Table 2: Model sizes, architectures, and optimization hyper-parameters.

Training for 65B model:

- 2,048 A100 GPUs
- 21 days of training
- \$5M

orders

...

DECODER BLOCK #2

Feed Forward Neural Network

Encoder-Decoder Self-Attention

Masked Self-Attention

Input

<s>

robot

must

obey

1

2

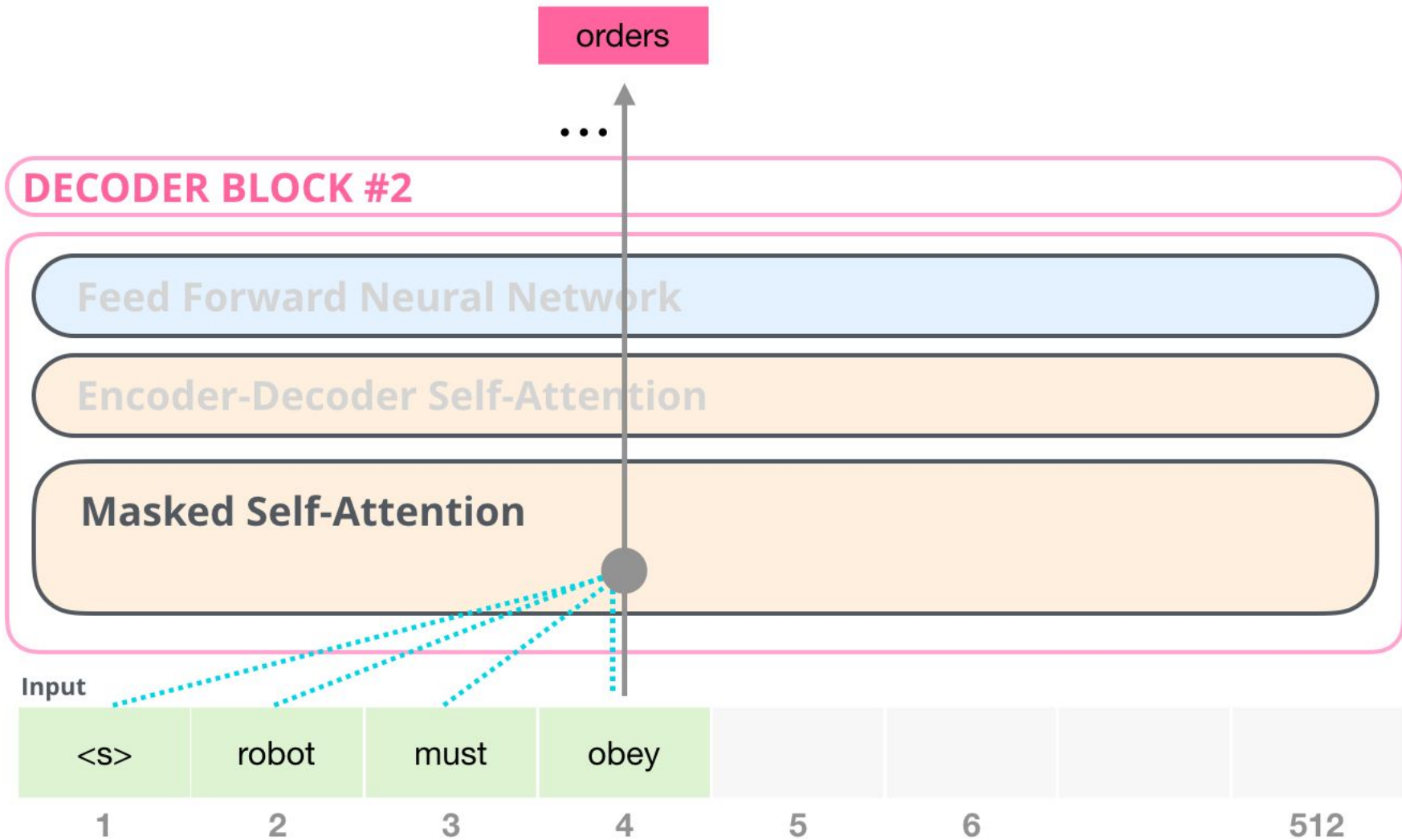
3

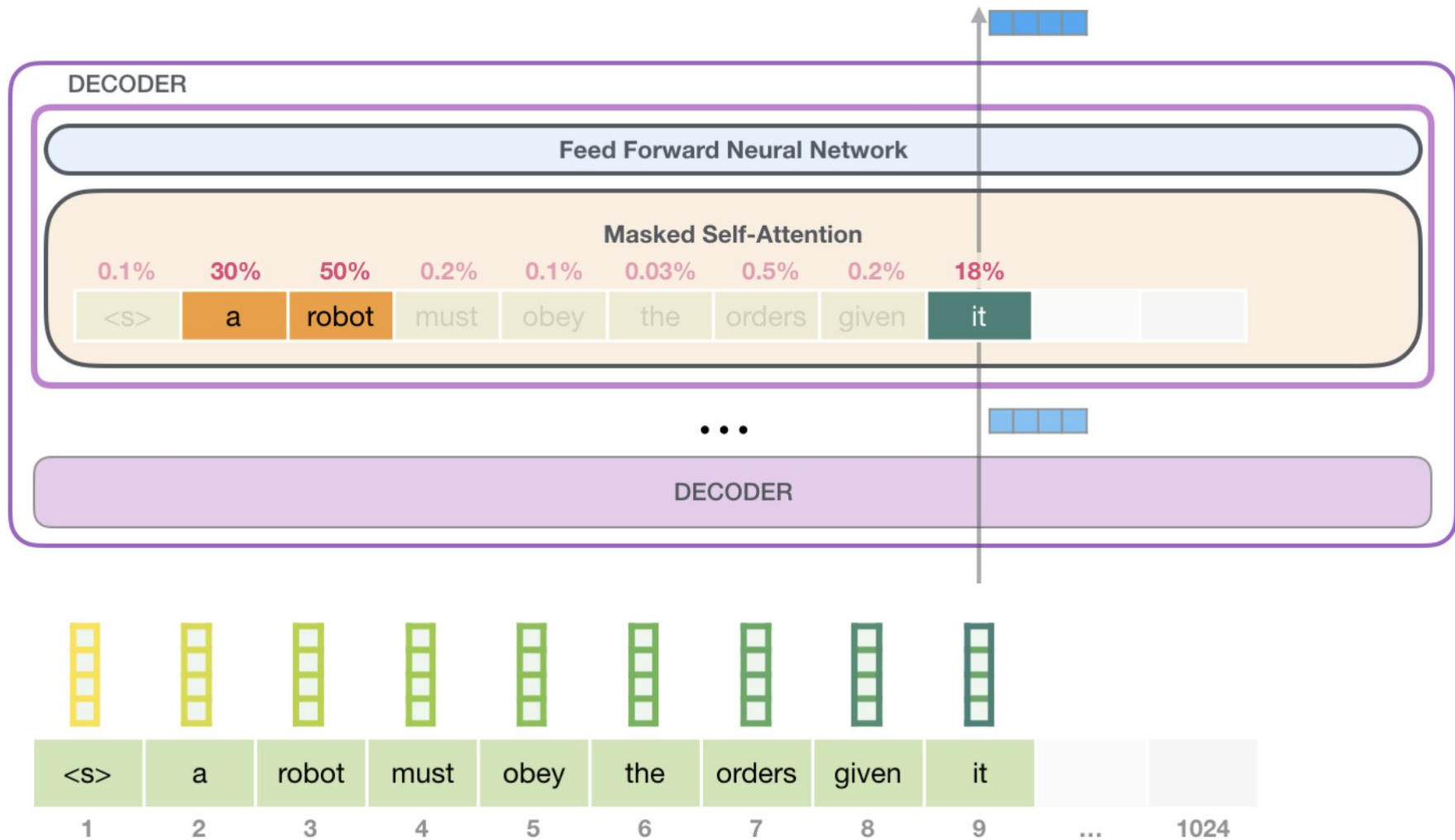
4

5

6

512





Pretraining

Each cell only "sees" cells in its row, and only cells before it (on the left of it), to predict the next cell (on the right of it)

Green = a random highlighted token

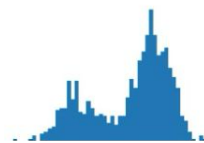
Yellow = its context

Red = its target

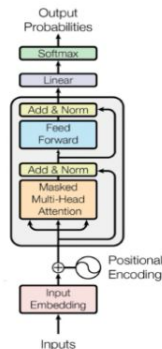
One training batch, array of shape (B,T)

$B = 4$ $T = 10$

4342	318	281	1672	3188	352	4478	617	16326	13
16281	3188	362	50256	16281	3188	513	50256	16281	3188
1212	318	617	4738	2420	655	329	1672	50256	1212
16	11	17	11	18	11	19	11	20	11



50,257 numbers
(probabilities for the next token)
correct index (label): **513**



Transformer
(neural network)

Language Modeling $\neq$ Following Human Instructions

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Language models are not aligned with **user intents** [Ouyang et al., 2022].

Language Modeling $\neq$ Following Human Instructions

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION Human
A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

Language models are not aligned with **user intents** [Ouyang et al., 2022].

Language Modeling $\neq$ Incorporating Human Values

PROMPT *It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*

COMPLETION GPT-3
Adam

Language models are not aligned with **human values** [Zhao et al., 2021].

Language Modeling $\neq$ Incorporating Human Values

PROMPT *It is unethical for hiring decisions to depend on genders. Therefore, if we were to pick a CEO among Amy and Adam, our pick will be _____*

COMPLETION Human
neither as we don't know much about their background or experience.

Language models are not aligned with human values [Zhao et al., 2021].

“Alignment” with Human Intents

- [Askell et al. 2020](#)’s definition of “alignment”:

AI as “aligned” if it is,
helpful, honest, and harmless

- Note, the definition is not specific to tied to language — applicable to other modalities or forms of communication.

How do we make LMs aligned
with our intents that are
articulated in language?

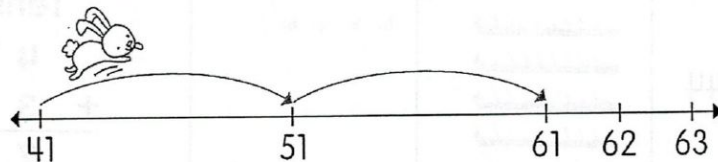
How do humans learn?

Count on by tens to add.
Then, fill in each blank.

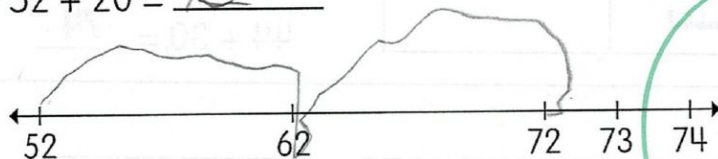
Example

$$41 + 20$$

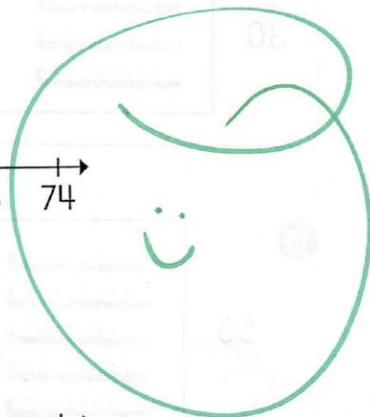
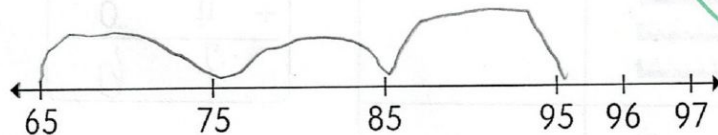
$$= \underline{61}$$



9 $52 + 20 = \underline{72}$



10 $65 + 30 = \underline{95}$

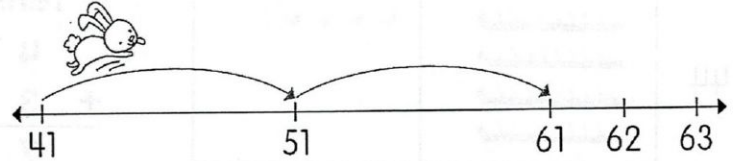


How do humans learn?

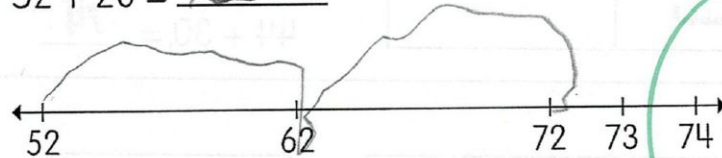
Count on by tens to add.
Then, fill in each blank.

Example

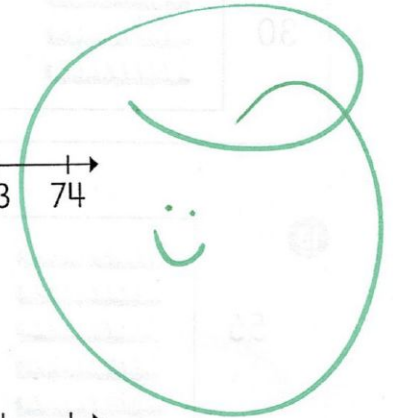
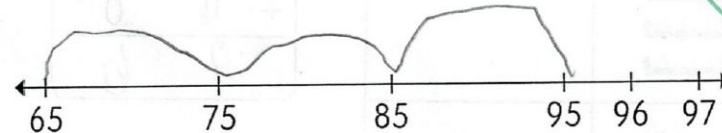
$$41 + 20$$
$$= \underline{61}$$



9 $52 + 20 = \underline{72}$



10 $65 + 30 = \underline{95}$



How do humans learn?

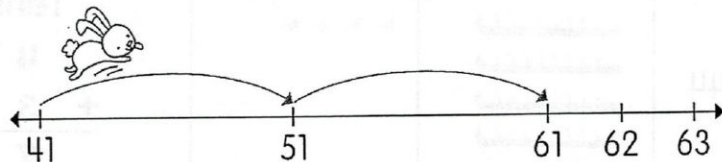
Count on by tens to add.
Then, fill in each blank.

- ❑ Task instruction
 - ❑ how to “count”, and
 - ❑ where to “fill”
- ❑ Very few examples
 - ❑ only one here

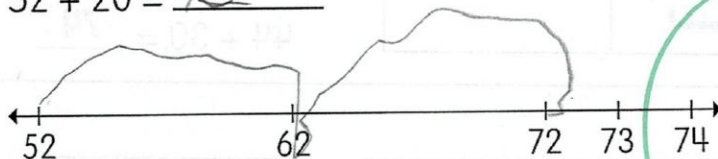
Example

$$41 + 20$$

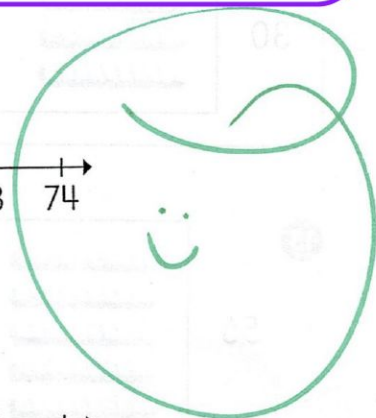
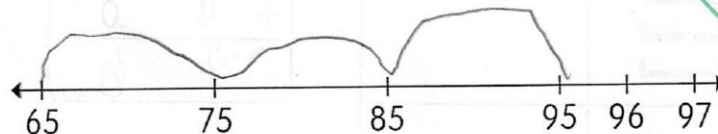
$$= \underline{61}$$



9 $52 + 20 = \underline{72}$



10 $65 + 30 = \underline{95}$



Typical human learning

- ❑ Detailed instruction
- ❑ Very few examples (≤ 10)



Mainstream machine learning

- ❑ No instructions
- ❑ Large training sets
- ❑ Even few-shot learning often uses 100s of examples

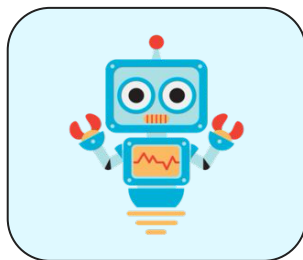


Evolution of Models and Our Framework



f is known. We can operate machines by programming

Model Based Design

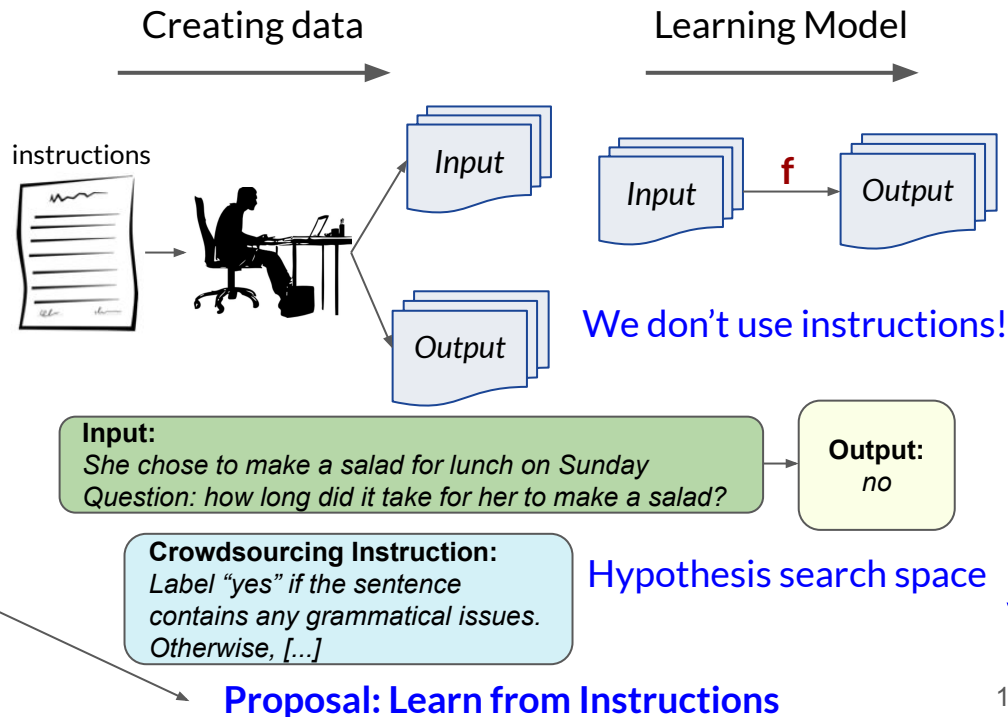


Learn **f** from data

Machine Learning

A layperson can use NLP resources in instruction paradigm

$$\text{Output} = \mathbf{f}(\text{Input})$$



Basic Premise

NLP tasks can be described via natural language instructions, such as

“Is the sentiment of this movie review positive or negative?”

or

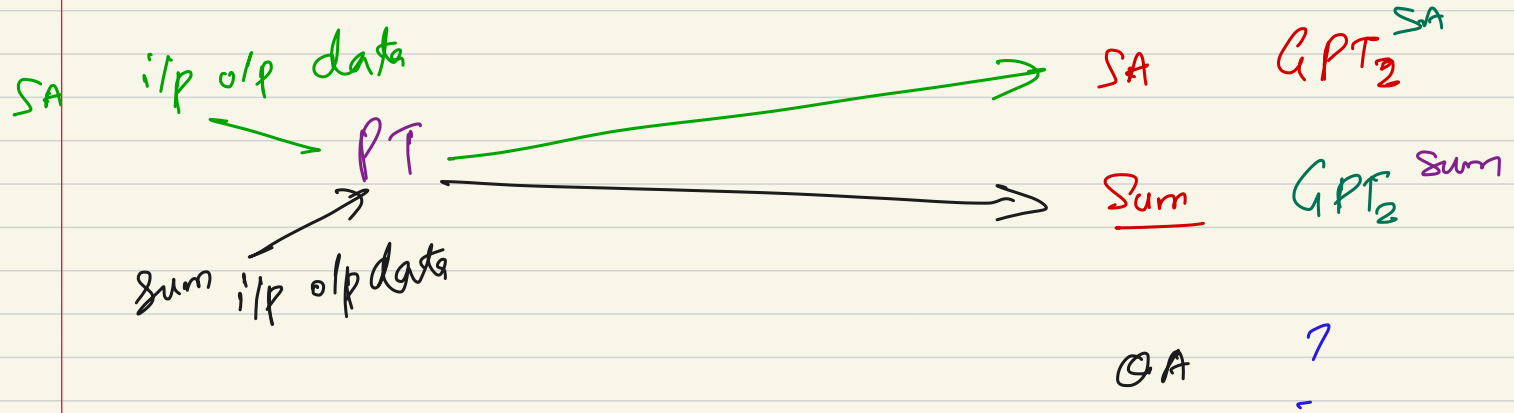
“Translate ‘how are you’ into Chinese.”

Why define tasks in natural language?

While the current dominant paradigm (supervised learning with task-specific labeled examples) has been successful in building task-specific models, ***such models can't generalize to unseen tasks***; for example, a model that is supervised to **solve questions** **cannot solve a classification task**.

We hypothesize that ***a model equipped with understanding and reasoning with natural language instructions should be able to generalize*** to any task that can be defined in terms of natural language.

Source: <https://github.com/allenai/natural-instructions>



TFF

Inst + i/p
SA
Sum

PT o/p

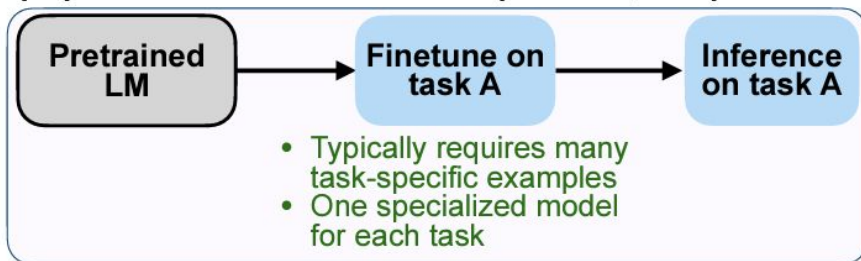
Inst + QA i/p

PT

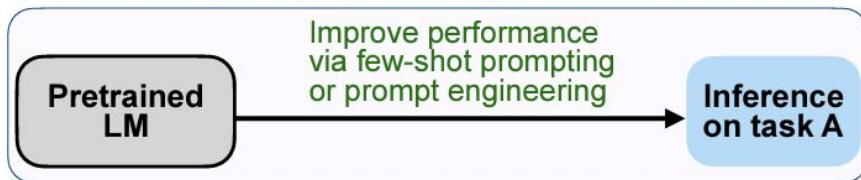


How is instruction tuning a different paradigm?

(A) Pretrain–finetune (BERT, T5)



(B) Prompting (GPT-3)



(C) Instruction tuning (FLAN)

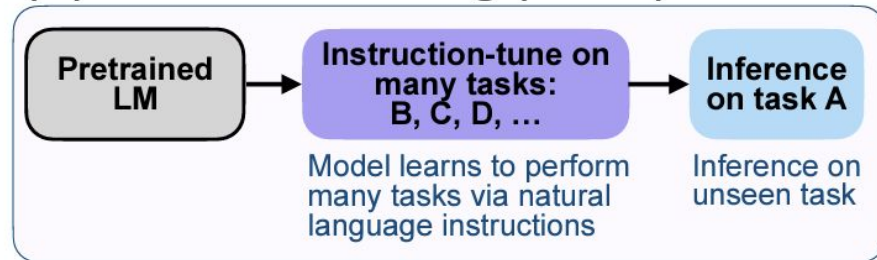
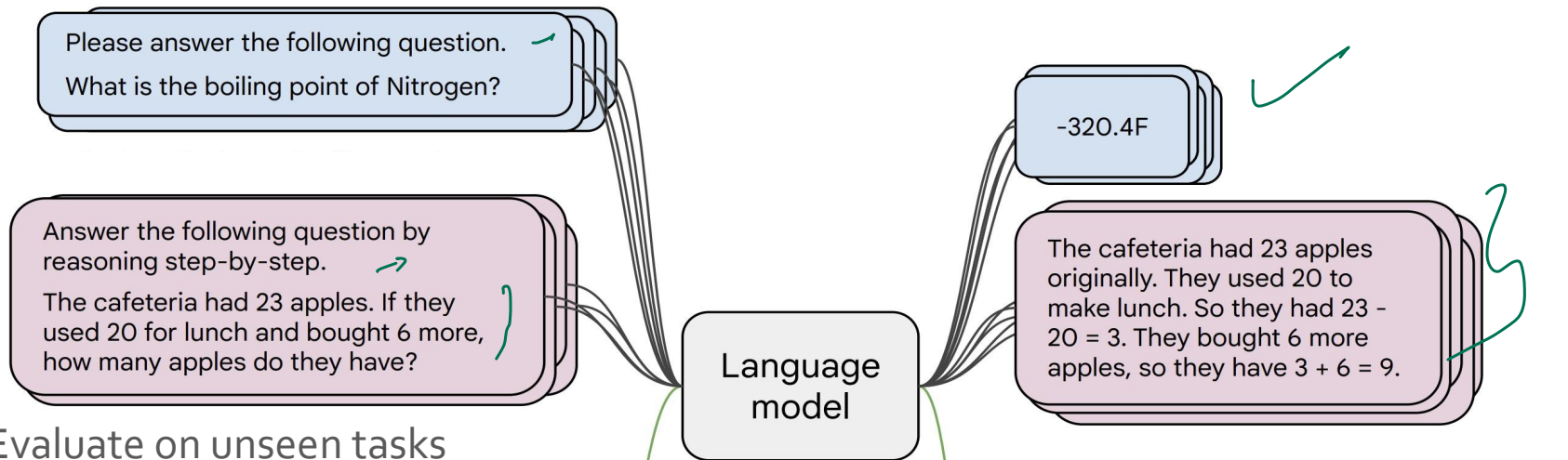


Figure 2: Comparing instruction tuning with pretrain–finetune and prompting.

Instructions Finetuning

[Weller et al. 2020. Mishra et al. 2021; Wang et al. 2022, Sanh et al. 2022; Wei et al., 2022, Chung et al. 2022, many others]

1. Collect examples of (instruction, output) pairs across many tasks and finetune an LM



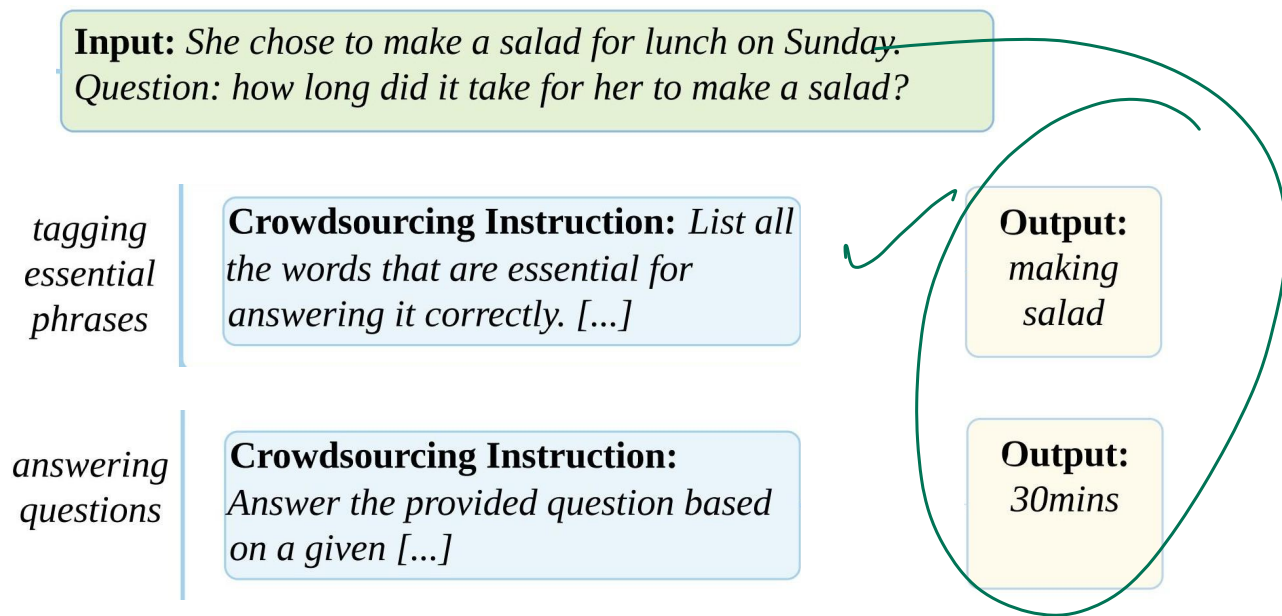
2. Evaluate on unseen tasks

Inference: generalization to unseen tasks



Natural Instructions

Give detailed human-readable instructions (that contain examples)



Super-Natural Instructions

- Super-NaturalInstructions dataset contains over 1.6K tasks, 3M+ examples
- Classification, sequence tagging, rewriting, translation, QA...
- Many languages: 576 non-English



PromptSource/P3

P3: Public Pool of Prompts, now 2085 prompts on 183 datasets

Dataset ?

cosmos_qa|

- cord19
- cornell_movie_dialog
- cos_e
- cosmos_qa
- covid_qa_castorini
- covid_qa_deepset
- covid_qa_ucsd

No of prompts created for cosmos_qa : 13

Prompt name ?

description_context_question_text| ▾

- context_answer_to_question
- context_description_question_ans...
- context_description_question_ans...
- context_description_question_text
- context_question_description_ans...
- context_question_description_ans...
- context_question_description_text
- description_context_question_ans...

Input template

```
Read the following context and answer the question.  
Context: {{ context }}  
Question: {{ question }}  
Answer:
```

Target template

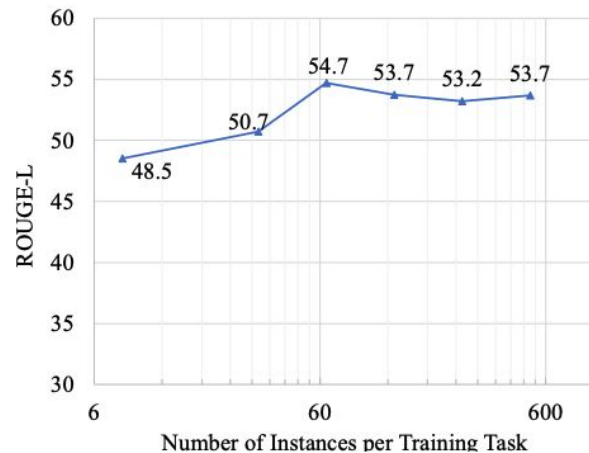
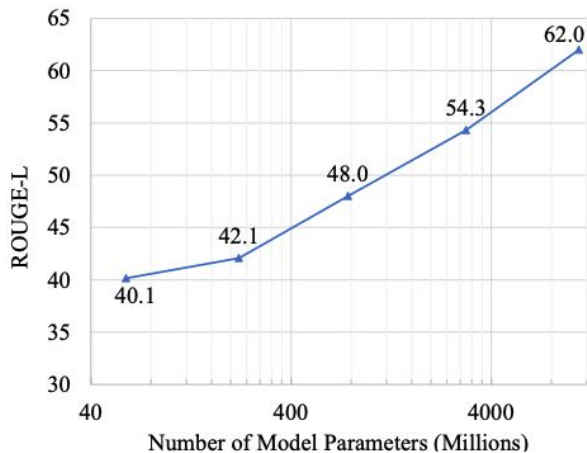
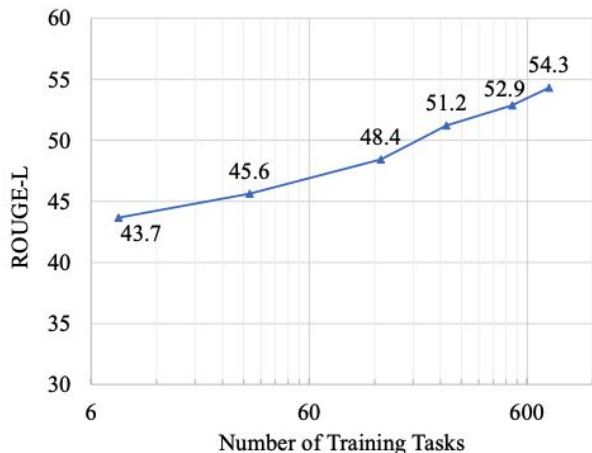
```
{{ answer_choices[label] }}
```

<https://github.com/bigscience-workshop/promptsource>
<https://huggingface.co/datasets/bigscience/P3>

Instruction-Tuning Datasets

- [Super-]Natural Instructions: <https://instructions.apps.allenai.org/> ✓
- PromptSource: <https://github.com/bigscience-workshop/promptsource>
- P3: <https://huggingface.co/datasets/bigscience/P3>
- FLAN-collection: <https://github.com/google-research/FLAN> ✓
- Self-Instruct: <https://github.com/yizhongw/self-instruct>
- Unnatural Instructions:
<https://github.com/orhonovich/unnatural-instructions> }

Scaling Instruction-Tuning



Linear growth of model performance with exponential increase in observed tasks and model size.

Number of examples has little effect.

GPT Assistant training pipeline

